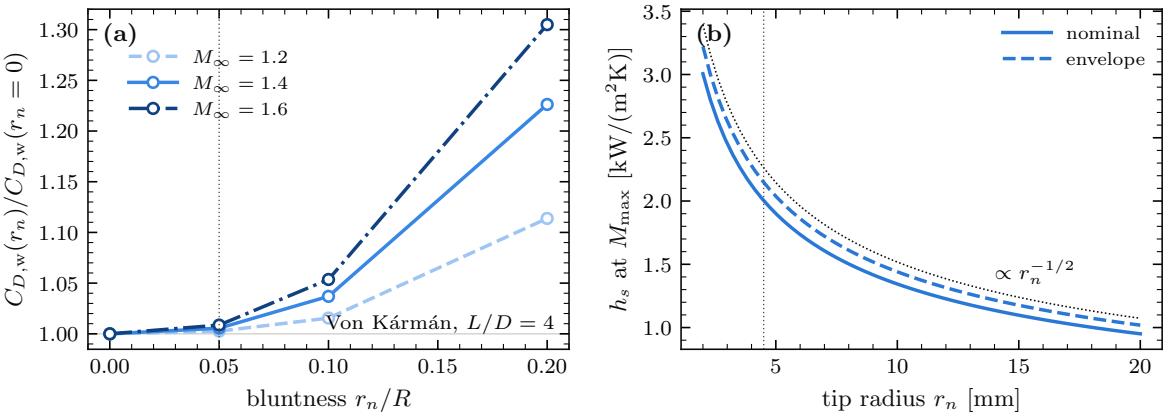

(a)
$M_\infty = 1.2$
$M_\infty = 1.4$
$M_\infty = 1.6$
Von Kármán, $L/D = 4$
$C_{D,\mathrm{w}}(r_n)/C_{D,\mathrm{w}}(r_n = 0)$
bluntness r_n/R
1.00
1.05
1.10
1.15
1.20
1.25
1.30
0.00
0.05
0.10
0.15
0.20
(b)
nominal
envelope
$\propto r_n^{-1/2}$
h_s at $M_{\max}$ [kW/(m^{2}K)]
tip radius r_n [mm]
1.0
1.5
2.0
2.5
3.0
3.5
5
10
15
20